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Black Women's Cambridge

A Case Study on Critical GIS and Black Feminist Geographies

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Inserting Black feminist perspectives into critical GIS

How might visualizing Black women's geographies disrupt traditional GIS practices?

Background

Black women's geographies describe how Black women's uniquely intersectional experience of racial-sexual hierarchies and oppression shape our creation and definition of geographic landscapes.

This creates an ambiguous, **subjective experience of geography** that captures communities made “both *for* [us] through processes of control and *by* [us] through means of resistance.”

Geographic information systems (GIS) were developed as an objective scientific tool without consideration of social, emotional, or cultural constructions of space.

In a field referred to as **critical GIS**, “Feminist [and Black] geographers challenge objectivity and mind/body dualisms by recognizing emotion and embodiment as ways of knowing,” emphasizing qualitative data, and rejecting simplification or generalization imbued in GIS features.

However, Black women's intersectional geographies and ways of knowing have not been directly considered or applied to critical GIS

A **research-through-design**
study of Black feminist
geographies using **oral histories**
and ArcGIS to visualize Black
women's geographies of
Cambridge.

Methods: Oral History Analysis

Common Neighborhoods, Places, and Streets

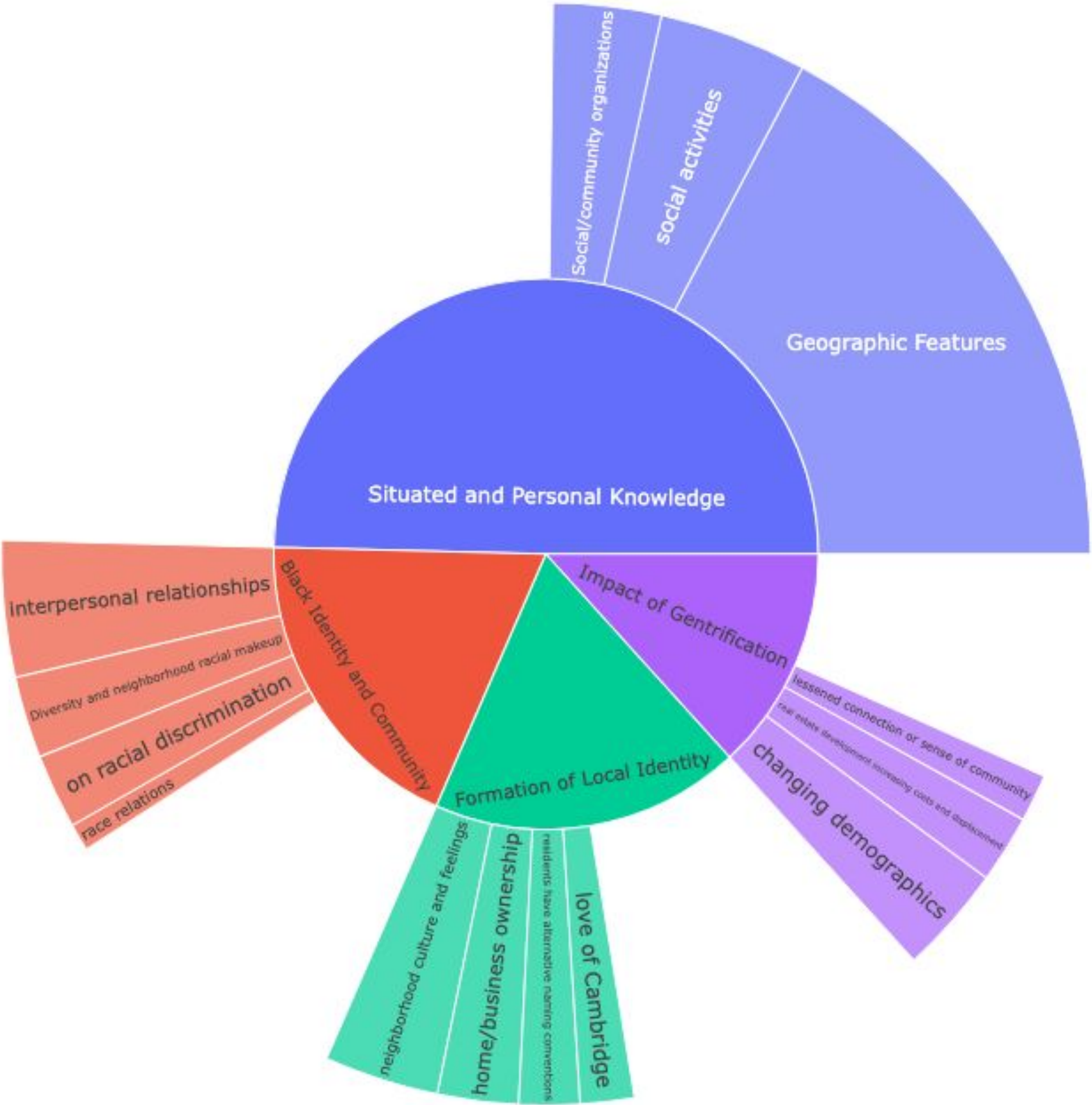
Names of churches, schools, streets, and other activity spaces to serve as anchors around which to orient more situated knowledge

Pride in Community and Homeownership

All women spoke with pride, affection, and nostalgia for their neighborhoods, particularly about homeownership and communal care

Prevalent Negativity about Gentrification

Despite conflicting opinions on race in the 20th century, all women mentioned demographic, social and cultural changes resulting from displacement caused by real estate speculation



Methods: Design Documentation



Design Diary: Black Women's Cambridge

This is a design diary documenting my research-through-design process for the ArcGIS map/website on Black women's geographies in Cambridge. These entries will be used to write the **results** and **conclusion** sections of the poster. The results will be a few major design phases (to be determined reflexively at the end), decisions or recurring instances of feminist data viz principles being applied, and a brief description of this log. The **conclusion** will then explain how the final prototype adheres to the overall project goals.

Sources

[Schedule of Documentation](#)

[Reflective Experiment Log](#)

[Template Block](#)

[7/10/26: Oral History Initial Review](#)

[7/13/26: GIS Exploration & Digital Collections Search](#)

[7/14/26: Oral History - Familiarization](#)

7/23/26 - Keywords

Houghton Elementary School The Coast "the city designates as River's side" Cambridge Community Center Regina Key 216 Green Street Pearl and Green St "East of Living" Central Square "the demographics have changed tremendously" Illness resource allocation advocacy service	Born on Pine Street several Homeowners in family Lived on Windsor Street Roberts School Margaret Fuller House - "the place to be" Cherry Street Neighborhood House "church was our entertainment" Barbados "A lot of people from Barbados in Cambridge" First Helena's Church Mrs. Nichols "called [churches] by their numbers" Multiracial Stukerian Apothecary on Hampshire Street Mother's Club "I don't know if there is a Black community" "All of my friends sold their houses"	Peabody Grammar School "people owned their own homes" Copley Street Fayerweather Street "I didn't feel anything racially different" Filing's Basement "I don't have that connection anymore" Ms. Jackson Concord Ave Cambridge High and Latin Concord Ave Saint Bartholomew's Church on Harvard St Cotillion The Peails North Cambridge (technically with Cambridge) House parties fresh Pond	Dimmock St, Boston. New England Hospital for Women and Children 211 Horry Avenue, 103 Inman Street, 306 Weldon Street Peabody on Linnaean St The Coast dwindling Black Community Church Sunday school "always welcome" "owned these houses" Fresh pond North Cambridge strawberry Hill Union Baptist Church "didn't feel any racism" price of property rising Brigham's "corner of...." ice skating Concord Avenue "technically it's called west Cambridge" Porter Square Mass Avenue
"identified... by our politicians" Margaret Fuller House "was once very diverse" The Port/ Area 4 (Federal designation) Inner Belt Pisani Center Washington Elms "Gentrification is done" Washington St Tensions between Barbadian and African Americans Central Square Godde's Bra Factory on the corner of Main and Windsor St "I don't know if there's anything like Cambridge"	Concord Ave Granville Road Fayerweather Street Huron Ave Hubbard Ave "racism was there" uplifting women "more people, I don't know" Community Barber Shop ↳ 726A Concord Ave "they were calling it North Cambridge" "once you cross Wessal Lane, everybody was white" "turned into West Cambridge" The Russel School Fresh Pond Christ Church Francis "Queenie" Pierce "I don't think they have any ties" "corner of Missouri"	Chicken salad sandwiches The Breakfast Club Cambridge Women's Community club "Not directly and not blatant" racism "You found people through whatever church you went to" Service mentorship family Concord Avenue AKA North Cambridge AKA West Cambridge AKA Huron village lessened sense of community "we're not going anywhere" Peabody School on Linnaean St "subtle" "microaggression" "who are all the schools named from?" Regina Key at the Houghton School Concord Avenue and Alpine Street	Mass Ave Baptist Church on Hampshire Street "knew the neighbors" "you got to know people" Linnaean Street walking to/from school Porter Square Whelan Park "invisible at school" Track meets Putnam Avenue, Green Street West Canton Street Southwest Corridor 186 Fayerweather Street very white Peabody Grammar School Russel Grammar School Brownies Huron Ave Elliot Ballroom Church Activities Copley Street House Parties coming together at the high school

Design Process: Data Feminism and Geospatial Technologies

Content – Geospatial Data

- **Consider Context:** Situated knowledge and personal geographies
- **Rethink Binaries:** Uplift endonyms expressed by Black women rather than exonyms designated by those in power
- **Embrace Pluralism:** Contradictory and complementary accounts of spatial and social phenomena

Form – Map Design

- **Embrace Pluralism:** Multiple points of entry and forms of information. Leveraging multimedia affordances
- **Embodiment and Affect:** Color choices associated with expressed or inferred emotions
- **Rethink Binaries:** Alternative conceptions of open/closed borders.

Process – Mapping Workflow

- **Reflexivity:** Using the documentation process to stop and reflect on my decisions given the purpose, medium and audience
- **Make Labor Visible:** Highlight the labor of the archivists and community groups recording the histories
- **Examine Power:**

Next Steps

Having completed the thematic analysis, I can begin the mapping process.

I will insert quotes and/or audio clips from the oral histories and refer back to them during the design process as needed.

Tuesday

- Storyboarding

Wednesday

- Iteration 1
- Solicit feedback and/or compare with literature

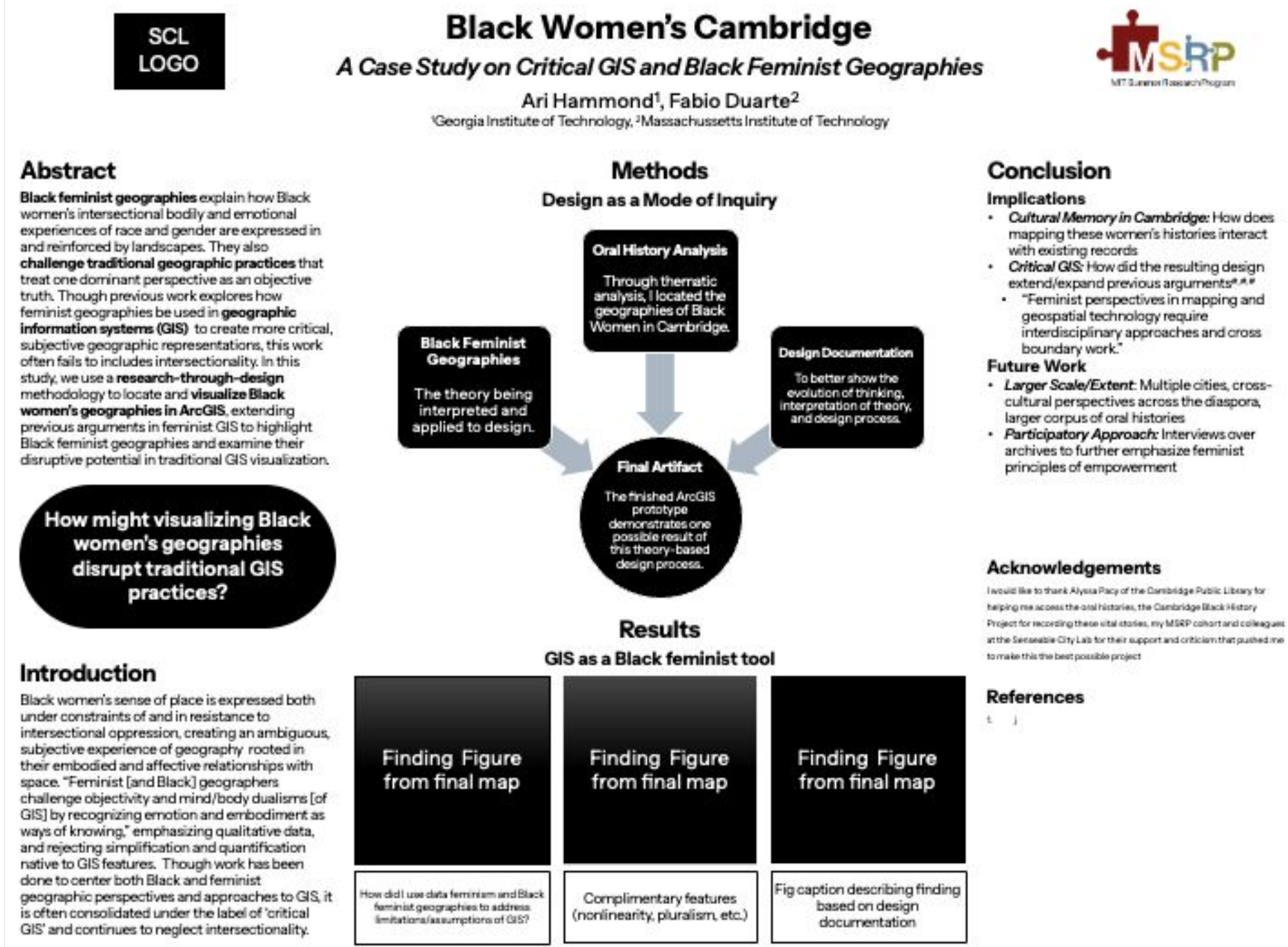
Thursday and Friday

- Revise to Final Iteration
- Complete poster

Current Poster Draft

To complete:

- Findings
 - Reviewing documentation and final map to reveal new design patterns in GIS
- Conclusion
 - Discuss implication for City of Cambridge
 - Explain impact to critical GIS
 - Future Work
- Selected references
 - 10-12 sources used for theory, methods, and oral histories



Black Women's Cambridge

A Case Study on Critical GIS and Black Feminist Geographies

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Abstract

Black feminist geographies explain how Black women's intersectional bodily and emotional experiences of race and gender are expressed in and reinforced by landscapes. They also **challenge traditional geographic practices** that treat one dominant perspective as an objective truth. Though previous work explores how feminist geographies be used in **geographic information systems (GIS)** to create more critical, subjective geographic representations, this work often fails to include intersectionality. In this study, we use a **research-through-design** methodology to locate and **visualize Black women's geographies in ArcGIS**, extending previous arguments in feminist GIS to highlight Black feminist geographies and examine their disruptive potential in traditional GIS visualization.

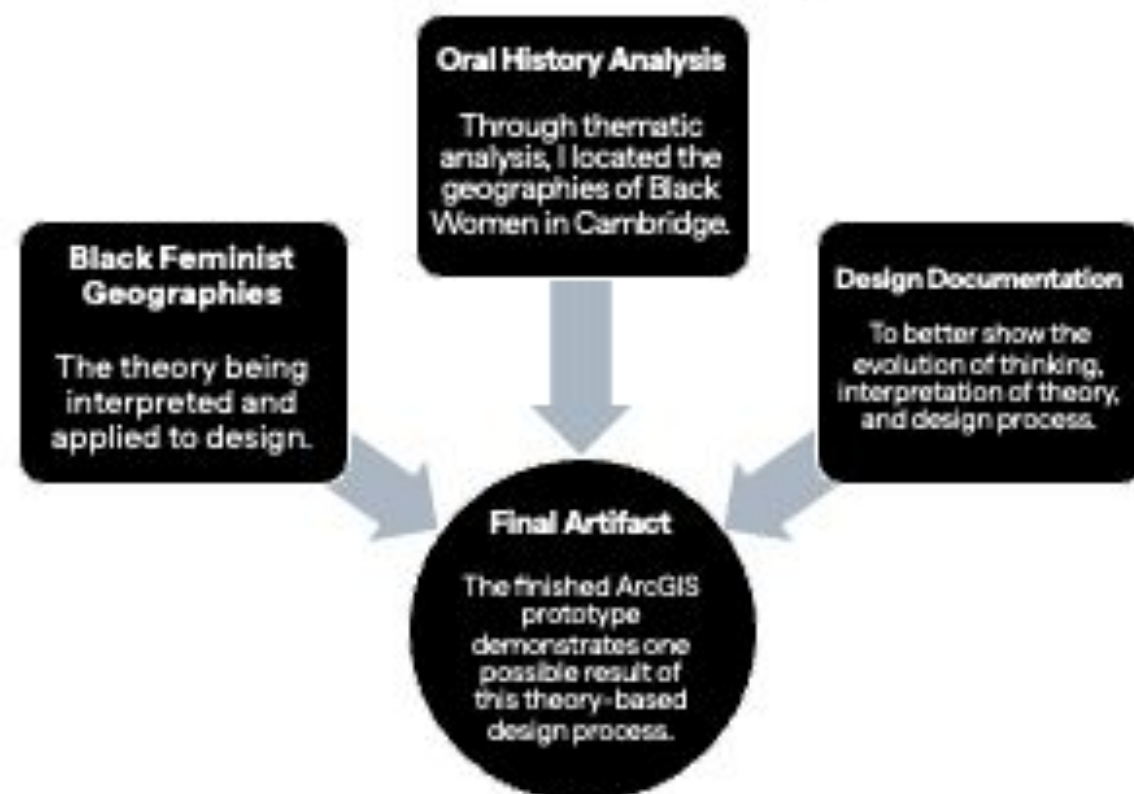
How might visualizing Black women's geographies disrupt traditional GIS practices?

Introduction

Black women's sense of place is expressed both under constraints of and in resistance to intersectional oppression, creating an ambiguous, subjective experience of geography rooted in their embodied and affective relationships with space. "Feminist [and Black] geographers challenge objectivity and mind/body dualisms [of GIS] by recognizing emotion and embodiment as ways of knowing," emphasizing qualitative data, and rejecting simplification and quantification native to GIS features. Though work has been done to center both Black and feminist geographic perspectives and approaches to GIS, it is often consolidated under the label of 'critical GIS' and continues to neglect intersectionality.

Methods

Design as a Mode of Inquiry



Results

GIS as a Black feminist tool

Finding Figure from final map	Finding Figure from final map	Finding Figure from final map
How did I use data feminism and Black feminist geographies to address limitations/assumptions of GIS?	Complimentary features (nonlinearity, pluralism, etc.)	Fig caption describing finding based on design documentation

Conclusion

Implications

- **Cultural Memory in Cambridge:** How does mapping these women's histories interact with existing records
- **Critical GIS:** How did the resulting design extend/expand previous arguments^{1,2}
- "Feminist perspectives in mapping and geospatial technology require interdisciplinary approaches and cross boundary work."

Future Work

- **Larger Scale/Extent:** Multiple cities, cross-cultural perspectives across the diaspora, larger corpus of oral histories
- **Participatory Approach:** Interviews over archives to further emphasize feminist principles of empowerment

Acknowledgements

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References

1. 2.